



Programme: Data Science and Business Analytics

CALCULUS

Lecture 3 (21.09.2026)

Properties of convergent sequences. Weierstrass theorem.

This lecture is about some important properties of convergent sequences and their properties. They are formulated in theorems 1 – 4.

1. Properties of convergent sequences.

Theorem 1 (on boundedness of convergent sequences). If $\{a_n\}$ is convergent then it is bounded sequence.

Proof. Assume that

$$\lim_{n \rightarrow \infty} a_n = a.$$

Then for $\varepsilon = 1$

$$\exists N \forall n > N : |a_n - a| < 1,$$

so that

$$|a_n| = |a_n - a + a| \leq |a_n - a| + |a| < |a| + 1 \quad \forall n > N.$$

Let us define

$$A = \max \{|a_1|, |a_2|, \dots, |a_N|, |a| + 1\}.$$

We obtain that $|a_n| \leq A$ for any $n \in \mathbb{N}$ and $\{a_n\}$ is bounded. $\square$

Theorem 2 (uniqueness of the limit of a convergent sequence). Any convergent sequence $\{a_n\}$ has the unique limit.

Proof. Assume the opposite, that is, there are real numbers $a \neq b$ such that

$$\lim_{n \rightarrow \infty} a_n = a, \quad \lim_{n \rightarrow \infty} a_n = b.$$

We can present $\{a_n\}$ in the form

$$a_n = \alpha_n + a, \quad a_n = \beta_n + b$$

where $\{\alpha_n\}$ and $\{\beta_n\}$ are infinitesimal. So, the sequence $\{\gamma_n\}$ with

$$\gamma_n = \alpha_n - \beta_n = (a_n - a) - (a_n - b) = b - a$$

is also infinitely small as a difference of two infinitesimal sequences. So, we have that $\lim_{n \rightarrow \infty} \gamma_n = 0$ which contradicts that $b - a \neq 0$. $\square$

Theorem 3 (arithmetic properties of convergent sequences). If $\{a_n\}$ and $\{b_n\}$ are convergent and

$$\lim_{n \rightarrow \infty} a_n = a, \quad \lim_{n \rightarrow \infty} b_n = b$$

then their sum $\{a_n \pm b_n\}$ and product $\{a_n \cdot b_n\}$ are also convergent sequences and

$$(a) \lim_{n \rightarrow \infty} (a_n \pm b_n) = a \pm b, \quad (b) \lim_{n \rightarrow \infty} (a_n \cdot b_n) = ab.$$

Moreover, if $b \neq 0$ and $b_n \neq 0$ for any $n \in \mathbb{N}$ then $\left\{\frac{a_n}{b_n}\right\}$ converges and

$$(c) \lim_{n \rightarrow \infty} \frac{a_n}{b_n} = \frac{a}{b}.$$

Proof. To prove (a) it is enough to remark that both $\{a_n\}$ and $\{b_n\}$ can be presented in the form

$$a_n = \alpha_n + a, \quad b_n = \beta_n + b \tag{1}$$

where $\{\alpha_n\}$ and $\{\beta_n\}$ are infinitesimal. So,

$$a_n \pm b_n = \alpha_n \pm \beta_n + a \pm b$$

and

$$(a_n \pm b_n) - (a \pm b) = \alpha_n \pm \beta_n$$

that is $\{a_n \pm b_n - (a \pm b)\}$ are infinitesimal, that is, convergent. Part (a) also follows.

To prove (b) from (1) we can find that

$$a_n \cdot b_n = (\alpha_n + a)(\beta_n + b) = ab + b\alpha_n + a\beta_n + \alpha_n\beta_n,$$

that is

$$a_n b_n - ab = b\alpha_n + a\beta_n + \alpha_n\beta_n.$$

So, $\{a_n b_n - ab\}$ is infinitesimal, that is, convergent, and we obtain part (b).

To prove (c) let us remark that

$$\frac{a_n}{b_n} - \frac{a}{b} = \frac{b \cdot a_n - a \cdot b_n}{b \cdot b_n} = \frac{b(\alpha_n + a) - a(\beta_n + b)}{b \cdot b_n} = \frac{1}{b_n} \left(\alpha_n - \frac{a}{b} \beta_n \right), \tag{2}$$

The further proof is based on the following

Lemma. If a convergent sequence $\{b_n\}$ such that $\lim_{n \rightarrow \infty} b_n = b$ satisfies the conditions $b \neq 0$ and $b_n \neq 0$ for any $n \in \mathbb{N}$, then the sequence $\left\{\frac{1}{b_n}\right\}$ is bounded.

Proof. Indeed, as $\lim_{n \rightarrow \infty} b_n = b \neq 0$ for $\varepsilon = |b|/2 \exists N \in \mathbb{N} \forall n > N : |b_n - b| < |b|/2$. So,

$$|b_n| = |b - (b - b_n)| \geq |b| - |b_n - b| > |b|/2,$$

and $\forall n > N : |b_n| > |b|/2$, or,

$$\left| \frac{1}{b_n} \right| < \frac{2}{|b|}.$$

If we take

$$A = \max \left\{ \left| \frac{1}{b_1} \right|, \left| \frac{1}{b_2} \right|, \dots, \left| \frac{1}{b_n} \right|, \frac{2}{|b|} \right\}$$

we get that the sequence $\left\{\frac{1}{b_n}\right\}$ is bounded by A . □

Returning to (2) we obtain that the sequence $\left\{ \frac{a_n}{a_n} - \frac{a}{b} \right\}$ is infinitesimal as the product of bounded and infinitesimal sequences. $\square$

Theorem 4 (sandwich theorem). *Assume that for any $n \in \mathbb{N}$ sequences $\{a_n\}$, $\{b_n\}$ and $\{c_n\}$ satisfy the inequality*

$$a_n \leq c_n \leq b_n. \quad (3)$$

If $\lim_{n \rightarrow \infty} a_n = \lim_{n \rightarrow \infty} b_n = a$ then $\lim_{n \rightarrow \infty} c_n = a$.

Proof. As $\lim_{n \rightarrow \infty} a_n = a$, by definition,

$$\forall \varepsilon > 0 \exists N_1 \forall n > N_1 : |a_n - a| < \varepsilon$$

and $a_n > a - \varepsilon$ for any $n > N_1$.

As $\lim_{n \rightarrow \infty} b_n = a$,

$$\forall \varepsilon > 0 \exists N_2 \forall n > N_2 : |b_n - a| < \varepsilon$$

and $b_n < a + \varepsilon$ for any $n > N_2$. Assume that $N = \max\{N_1, N_2\}$. Then $\forall n > N$

$$a - \varepsilon < a_n \leq b_n < a + \varepsilon,$$

or, by (3)

$$a - \varepsilon < a_n \leq c_n \leq b_n < a + \varepsilon$$

We get that

$$\forall \varepsilon > 0 \exists N \forall n > N : |c_n - a| < \varepsilon, \text{ that is } \lim_{n \rightarrow \infty} c_n = a.$$

Theorem follows. $\square$

2. Monotone sequences. Weierstrass theorem.

Definition 1. A sequence $\{a_n\}$ is called

- *increasing* if $a_{n+1} > a_n$ for any $n \in \mathbb{N}$;
- *non-decreasing* if $a_{n+1} \geq a_n$ for any $n \in \mathbb{N}$;
- *decreasing* if $a_{n+1} < a_n$ for any $n \in \mathbb{N}$;
- *non-increasing* if $a_{n+1} \leq a_n$ for any $n \in \mathbb{N}$.

Theorem 5 (Weierstrass theorem). *Any monotone bounded sequence $\{a_n\}$ has a limit.*

Proof. Without loss of generality, let us assume that the sequence $\{a_n\}$ is decreasing, that is, $a_{n+1} < a_n$ for any $n \in \mathbb{N}$. Let us consider the set X of members of the sequence

$$X = \{a_1, a_2, \dots, a_n, \dots\}$$

It is evident that X is not empty and bounded below. So, there is $\inf X = a$. Let us prove that $\lim_{n \rightarrow \infty} a_n = a$. Indeed, by definition of the infimum, $a_n \geq a$ for any $n \in \mathbb{N}$. Also, for any $\varepsilon > 0$ there is such $N \in \mathbb{N}$ that $a \leq a_N \leq a + \varepsilon$. As $\{a_n\}$ is decreasing all members of the sequence with $n > N$ satisfy the condition $a \leq a_n < a_N \leq a + \varepsilon$ so that $|a_n - a| < \varepsilon$. By definition of the limit, we obtain that $\lim_{n \rightarrow \infty} a_n = a$. For other types of monotone sequences the theorem can be proved analogously. $\square$